Multimodal Comfort-Aware Human–Robot Handover: A Real-Time Perception Pipeline for Adaptive Service Robot Interaction

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Abstract

Social and service robots performing object handovers must adapt to how comfortable each user feels; the same behavior can feel natural to one person and intrusive to another. We present a real-time comfort-scoring pipeline built on *late fusion of pretrained unimodal models*—facial emotion recognition (EfficientNet-B0/AffectNet), gaze estimation (L2CS-Net/Gaze360), and body posture analysis (MediaPipe Pose with depth-based withdrawal detection)—requiring no joint training. The fused system produces a continuous 0–100 comfort score from RGB-D video at 13.6–58.2 FPS and is designed for few-shot optimization at the deployment site, where fusion weights and penalty parameters are calibrated on a small number of environment-specific recordings. On a custom-gathered dataset using an Intel RealSense D435 across six handover scenarios, multimodal fusion captures comfort transitions that no single modality can track alone—for example, depth-based withdrawal detection identifies discomfort that facial expressions alone miss, and gaze aversion compensates when facial emotion is ambiguous. We situate this pipeline within a two-phase interaction framework of intent signaling and service execution, where the comfort score governs robot abort decisions.

1 Introduction

Social and service robots are increasingly deployed in settings where they must hand objects to people. A critical challenge is that the same robot behavior can feel natural to one person yet intrusive to another. If robots cannot perceive and adapt to individual comfort in real time, they risk triggering avoidance or safety incidents, limiting broader HRI adoption.

Despite progress in multimodal perception and comfort measurement, prior work has not closed the full perception-to-action loop for socially adaptive service. Existing systems either rely on rule-based switching between discrete behavior modes or study emotion recognition in isolation from robot control. Key gaps remain: (1) no system fuses face, gaze, and posture into a single continuous comfort score at interactive frame rates; (2) individual modalities have known limitations—facial ambiguity under neutral expressions, gaze unavailability when the face is occluded, and missed withdrawal in 2D pose—yet no prior system combines all three with depth-based withdrawal detection for handover scenarios; and (3) robot adaptation has not been driven by such a fused signal.

We address these gaps by designing a multimodal comfort-scoring pipeline within a two-phase interaction framework—intent signaling followed by adaptive service execution—where the comfort

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score governs robot abort decisions. We detail the architectural choices, fusion strategy, and dataset design, and present observations of how multimodal fusion yields more reliable comfort estimation than any single modality.

2 Related Work

Prior work spans three areas: multimodal perception, comfort measurement, and affect-aware decision-making.

Multimodal perception. Churamani et al. [15] combine facial expression recognition and speech via late fusion into an Affective Core (GWR network) with DDPG policy learning—the closest pipeline to ours, though they target dialogue rather than handover and omit gaze. Surveys by Wang and Feng [2], Su et al. [3], and Zhao et al. [4] identify cognition-level comfort fusion, temporal alignment across modalities, and the perception-to-action loop as key underexplored gaps. Fusion approaches in the literature range from confidence-weighted attention [17] and bilevel NAS [18] to vision-language models such as PaLM-E [20].

Comfort measurement. Yan and Jia [5] propose a comfort taxonomy (ergonomic, robot-motion, anthropomorphism, sociability) with subjective and objective instruments; Yan et al. [6] map physical factors to comfort but omit emotional signals. Gonzalez-Santocildes et al. [8] demonstrate comfort-driven RL policy optimization, and Heinisch et al. [9] release AFFECT-HRI, the only public HRI dataset pairing physiological data with affect labels.

Affect-aware interaction. Alrefaie et al. [12] study a UR10 robot with normal/slow/stop switching based on proximity—the same graduated structure we adopt—though their controller is purely rule-based. Zheng et al. [13] and Pandya [14] ground decisions in POMDPs and Bayesian MDPs, while Brunke et al. [16] survey safe learning in robotics. Current work focuses on industrial ergonomics [7] or proxemic RL [8]; constrained ML and POMDPs are rarely synthesized for social interaction.

Summary. To our knowledge, no existing system fuses facial emotion, gaze, and body posture from RGB-D video into a continuous real-time comfort score for handover scenarios. Our work bridges this gap with a late-fusion pipeline situated within a two-phase interaction framework.

3 Problem Statement

Let $\mathbf{x}_t = (I_c^{(t)},\, I_d^{(t)})$ denote an RGB-D frame pair captured at time t during a handover interaction, where $I_c^{(t)}$ is the color frame and $I_d^{(t)}$ is the aligned depth map from an Intel RealSense D435. Our goal is to estimate a continuous comfort score $C_t \in [0, 100]$ that reflects the user's real-time affective state. We decompose C_t into two modality-specific sub-scores:

$$C_t = w_e \cdot C_e^{(t)} + w_p \cdot C_p^{(t)}, \tag{1}$$

where $C_e^{(t)} = f_{\text{emotion}}(I_c^{(t)};\, \theta_E,\, \theta_G)$ fuses facial emotion valence (predicted by an EfficientNet-B0 classifier parameterized by θ_E) with a gaze engagement signal (estimated by L2CS-Net parameterized by θ_G), and $C_p^{(t)} = f_{\text{posture}}(I_c^{(t)},\, I_d^{(t)})$ integrates body openness from MediaPipe Pose landmarks with a depth-based withdrawal detector that flags sudden increases in user-to-camera distance. Temporal smoothing is applied via an exponential moving average (EMA):

$$\tilde{C}_t = \alpha \cdot C_t + (1 - \alpha) \cdot \tilde{C}_{t-1}, \tag{2}$$

where α is the smoothing factor. The weighting $w_e = 0.55$, $w_p = 0.45$ reflects a slight prior toward the emotion channel, as facial signals tend to be more informative during the approach phase of a handover when postural changes are small. The objective is that $\tilde{C}_t$ should (i) correlate with ground-truth comfort annotations, (ii) run in real time ($\geq$ 13 FPS), and (iii) be robust across conditions where any single modality is ambiguous or unavailable.

4 Proposed Approach

Our system explores a real-time comfort-aware perception pipeline designed for human–robot handover scenarios. The system integrates multimodal data streams—visual and depth information—to compute a unified comfort score reflecting both emotional and physical states. The architecture consists of two main branches: the *Emotion Detection Branch* and the *Posture Detection Branch*, combined using a weighted late-fusion scoring mechanism.

4.1 Emotion Detection Branch

Face detection. Given an input color frame I_c, we detect a face bounding box B using RetinaFace:

$$B = \text{RetinaFace}(I_c). \tag{3}$$

Emotion classification. The detected face region I_f (cropped from I_c using B) is passed to an EfficientNet-B0 model pretrained on AffectNet [22]. The model produces an emotion prediction E:

$$E = \text{EfficientNet-B0}(I_f;\, \theta_E). \tag{4}$$

The classifier outputs a categorical emotion label alongside a valence score that is mapped to the emotion comfort sub-score.

Gaze estimation. The gaze direction $G = (\text{yaw},\, \text{pitch})$ is estimated using L2CS-Net [21], which was pretrained on the Gaze360 dataset:

$$G = \text{L2CS-Net}(I_c;\, \theta_G). \tag{5}$$

Gaze engagement is quantified as the angular deviation from direct eye contact; small deviations indicate engagement, while large aversions signal discomfort or distraction. A gaze weight $w_g = 0.6$ modulates the contribution of gaze to the emotion sub-score.

4.2 Posture Detection Branch

Pose estimation. Using MediaPipe Pose, the skeletal pose P (a set of 33 body landmarks) is extracted from the color frame:

$$P = \text{MediaPipePose}(I_c). \tag{6}$$

We compute a body openness score from the relative positions of key joints (shoulders, elbows, wrists). Closed postures (e.g., crossed arms, hunched shoulders) reduce the posture comfort sub-score, while open and relaxed postures increase it.

Depth-based withdrawal and mouth covering detection. The depth map I_d is used to detect two discomfort indicators. First, a *withdrawal detector* monitors the user's torso depth over time; a sudden increase in user-to-camera distance beyond a threshold flags a withdrawal event, applying a penalty of $\delta_w = 25.0$ to the posture comfort score. Second, a *mouth-covering detector* checks whether hand landmarks overlap with the mouth region in 3D space, applying a penalty of $\delta_m = 30.0$:

$$W = \text{DepthAnalysis}(I_d,\, P). \tag{7}$$

4.3 Late Fusion for Comfort Scoring

The overall comfort score C_t is computed as the weighted combination of the emotion comfort sub-score C_e and the posture comfort sub-score C_p, as defined in Eq. (1). Both sub-scores are individually normalized to the $[0, 100]$ range before fusion. The fused score is temporally smoothed via the EMA defined in Eq. (2) with smoothing factor $\alpha = 0.85$ to suppress frame-level noise while preserving meaningful comfort transitions.

This modular late-fusion design allows each branch to operate independently, enabling the system to gracefully degrade when one modality is unavailable (e.g., the face is occluded or the user is out of the depth sensor's range). Furthermore, because each unimodal model is pretrained and used without additional training, the system is deployable with zero fine-tuning overhead.

5 Experimental Methodology

5.1 Dataset

We collected 33 RGB-D recordings (`.bag` files) using an Intel RealSense D435, covering six handover scenarios: walk-by (8), comfortable handoff (7), gradual discomfort (5), sudden withdrawal (4), distracted (5), and hesitant (4). All recordings were used for qualitative evaluation and parameter tuning; no formal train/test split was applied at this stage.

5.2 Pretrained Models

The system leverages pretrained models without additional training:

- **RetinaFace**: face detection from color frames.
- **EfficientNet-B0**: emotion classification, pretrained on AffectNet [22].
- **L2CS-Net**: gaze direction estimation, pretrained on Gaze360 [21].
- **MediaPipe Pose**: 33-landmark skeletal pose detection.

5.3 Fusion Weight Calibration

The late-fusion strategy combines unimodal outputs into a unified comfort score. Calibration experiments were conducted to determine the optimal weights w_e and w_p for the emotion and posture branches by manually tuning across all recordings. Initial optimization targeted maximizing the comfort score on the successful handoff scenario (sc02) while minimizing it on all other scenarios.

5.4 Hyperparameters and Configuration

The system configuration used for all reported experiments is: $w_e = 0.55$, $w_p = 0.45$, gaze weight $w_g = 0.6$, mouth-covering penalty $\delta_m = 30.0$, withdrawal penalty $\delta_w = 25.0$, EMA smoothing factor $\alpha = 0.85$, and a posture gamma of $\gamma = 0.5$ with delta $\delta = 0.3$.

5.5 Evaluation Metrics

We evaluate the system along three dimensions:

- **Emotion detection**: accuracy and F1-score for categorical emotion classification.
- **Posture detection**: precision and recall for withdrawal and mouth-covering event detection.
- **Integrated comfort scorer**: correlation between computed comfort scores and expected scenario-level comfort rankings, along with detection rates (face detection rate, pose detection rate) and event rates (mouth-covering rate, withdrawal rate).

6 Results and Discussion

6.1 Scenario-Level Comfort Scores

Table 1 summarizes the aggregated metrics across all six scenarios. The "Integrated (mid)" column reports the blended comfort score during the handoff window (middle 50% of each recording), while "Emotion (mid)" and "Posture (mid)" report the individual branch contributions. "Peak" denotes the maximum integrated score observed.

6.2 Qualitative Results

Figure 1 shows representative frames from three scenarios with the system's real-time detection overlays, including facial emotion labels, gaze vectors, and skeletal pose landmarks.

6.3 Discussion

Multimodal fusion captures complementary signals. The comfortable handoff scenario (sc02) reliably produces the highest emotion comfort score (55.4) and the highest peak integrated score

Table 1: Scenario summary: comfort scores and detection rates across six handover scenarios. Scores are computed over the middle 50% of each recording (the handoff window).

Scenario	Files	Integ.(mid)	Emot.(mid)	Post.(mid)	Peak	Face%	Pose%	Mouth%	Wdraw%
sc01_walkby	8	55.7	46.9	66.6	73.9	15	33	8	2
sc02_comfortable	7	63.1	55.4	72.6	82.5	36	37	0	2
sc03_grad._discomfort	5	62.7	49.4	78.9	79.1	29	36	1	0
sc04_sudden_wdraw	4	58.8	49.7	70.0	81.9	27	30	1	0
sc05_distracted	5	63.2	51.0	78.1	76.1	27	35	3	0
sc06_hesitant	4	62.1	49.0	78.0	77.5	20	30	3	0

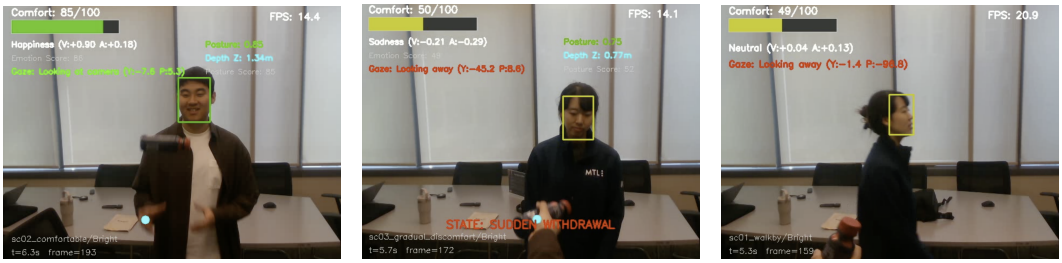

Figure 1: Representative frames from three handover scenarios. **Left**: Comfortable handoff (sc02)—relaxed posture and gaze engagement yield a high comfort score. **Center**: Sudden withdrawal (sc04)—the user pulls back, triggering the depth-based withdrawal detector. **Right**: Walk-by (sc01)—gaze aversion and minimal face detection produce the lowest comfort score.

(82.5), confirming that the fusion pipeline can distinguish positive interactions. Meanwhile, the walk-by scenario (sc01) yields the lowest integrated score (55.7) and the lowest emotion sub-score (46.9), consistent with the lack of user engagement. Notably, the posture branch provides higher scores across discomfort scenarios (sc03–sc06) than the emotion branch, suggesting that postural cues alone are insufficient to discriminate comfort levels—the emotion channel is critical for separating comfortable from uncomfortable interactions.

Detection rates highlight modality availability challenges. Face detection rates range from 15% (sc01_walkby) to 36% (sc02_comfortable), reflecting the fact that users who are not engaging with the robot are less likely to face the camera. Pose detection rates are somewhat higher (30–37%) but still limited. These low detection rates underscore the importance of graceful degradation in a multimodal system: when the face is unavailable, the posture branch must carry the comfort estimation, and vice versa.

Failure analysis and limitations. Several limitations emerged during our initial experiments:

1. **Temporal windowing.** The current system averages the comfort score across an entire recording, including periods when the user is walking toward the robot or walking away after a completed handoff. These non-interaction segments dilute the signal from the actual handoff window, skewing the average comfort scores. The "mid" scores in Table 1 (middle 50% of each recording) partially address this, but a principled start/end detection mechanism is needed.
2. **Gaze dynamics during handover.** We observed that gaze shifts from the robot to the item location once the user begins reaching, meaning gaze aversion during the handoff phase may not indicate discomfort but rather a natural attention shift. This suggests gaze may be most informative during the intent-signaling phase rather than throughout the entire interaction.
3. **Scenario granularity.** Six scenarios may introduce excessive variability with too few recordings per scenario (4–8). Consolidating into fewer, more distinct categories with more recordings each would improve statistical reliability and simplify the optimization landscape.
4. **Few-shot optimization.** Our initial attempt to optimize fusion weights by maximizing the comfort score on the comfortable scenario and minimizing it on all others did not converge, likely due to the small sample size and the overlap in comfort score distributions across

scenarios. The data collection methodology was not structured to support effective few-shot learning—recordings lacked precise temporal annotations for handoff start and end times.

7 Next Steps and New Research Ideas

Building on the insights and limitations from our midterm experiments, we outline several directions for the final report.

7.1 Refined Data Collection Protocol

Our most critical next step is redesigning the data collection methodology. We plan to introduce explicit temporal markers for the handoff window—defining the start (robot initiates approach) and end (object transferred or interaction aborted) of each interaction. This will eliminate the averaging artifacts from non-interaction segments and produce cleaner training signals for comfort score optimization. We also plan to reduce the number of scenarios from six to three or four core categories (e.g., comfortable, uncomfortable, and disengaged) while increasing the number of recordings per scenario to at least eight, improving statistical power. With the restructured dataset, we will record only four `.bag` files per scenario—three for training and one for testing—keeping data collection low-effort and aligned with our goal of simple, few-shot optimization of fusion weights and penalty parameters at the deployment site.

7.2 Phase-Aware Comfort Scoring

Motivated by the observation that gaze dynamics change across interaction phases, we plan to implement a phase-aware scoring strategy aligned with the two-phase framework from our proposal (Idea 3: Closed-Loop Framework [15]). During the *intent signaling* phase (robot approaches the user), gaze engagement would be weighted more heavily as an indicator of willingness and readiness. During the *service execution* phase (active handoff), the system would shift emphasis to posture and depth-based signals, since gaze naturally redirects toward the object. This phase-dependent reweighting can be formalized as:

$$C_t^{\text{phase}} = \begin{cases} w_e^{(\text{intent})} \cdot C_e^{(t)} + w_p^{(\text{intent})} \cdot C_p^{(t)}, & \text{if } t \in \text{intent signaling}, \\ w_e^{(\text{exec})} \cdot C_e^{(t)} + w_p^{(\text{exec})} \cdot C_p^{(t)}, & \text{if } t \in \text{service execution}, \end{cases} \tag{8}$$

where the intent-phase weights emphasize gaze (higher $w_e^{(\text{intent})}$) and the execution-phase weights emphasize posture and depth (higher $w_p^{(\text{exec})}$).

7.3 Closed-Loop Robot Integration

We have begun interfacing with the robot's API and plan to implement a binary decision policy driven by the comfort score $\tilde{C}_t$:

- **Continue** ($\tilde{C}_t >$ threshold): proceed with the handover at normal speed.
- **Abort** ($\tilde{C}_t \leq$ threshold or withdrawal detected): immediately retract and retreat to a safe distance.

We simplify the originally proposed three-tier state machine (continue/slow/abort) to a binary continue/abort decision to reduce scope and allow us to focus on reliably detecting the abort condition. This integration will close the perception-to-action loop, moving from offline comfort scoring to real-time adaptive behavior.

7.4 On-Site Generalization via Few-Shot Fine-Tuning

Following Idea 4 from our proposal, we plan to investigate whether a small, environment-specific calibration dataset (collected on-site with the deployment camera and lighting) can improve comfort classification accuracy compared to the purely pretrained pipeline. Specifically, we hypothesize that fine-tuning the fusion weights w_e and w_p (and potentially the gaze and penalty parameters) on 5–10 minutes of site-specific recordings will reduce false-stop triggers—premature aborts caused by

misclassified discomfort due to domain shift in lighting, camera angle, or user demographics. We plan to evaluate this through a within-subject comparison: the same set of test recordings scored by the generic pipeline versus the site-adapted pipeline, measuring improvement in comfort score separation between comfortable and uncomfortable scenarios.

A Robot Hardware Setup

In addition to developing the multimodal comfort-scoring pipeline, we spent this period getting familiar with the robot manipulator we plan to use for the final project. Figure 2 shows the arm during our initial bring-up sessions. This hands-on work included learning the robot's API, understanding its workspace and joint limits, and performing preliminary reach-and-retract motions that will form the basis of the handover trajectory. Going forward, this platform will serve as the deployment target for our late-fusion comfort model and for recording new interaction data with human participants.

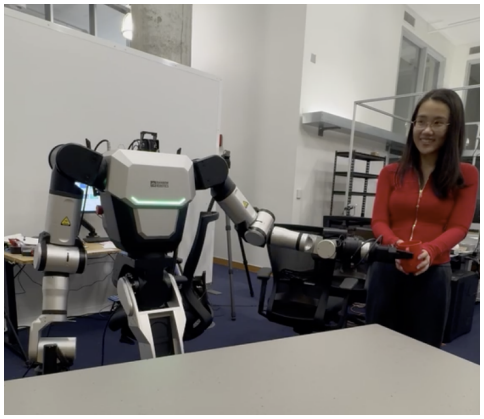

Figure 2: The robot manipulator used for our project. During this period we familiarized ourselves with its control API and workspace in preparation for deploying the comfort-aware perception pipeline and recording handover interaction data.

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